Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/10/2004
PI Name:	Davis, Stephen H. Ph.D.		
Project Title:	Interfacial Dynamics		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sdavis@northwestern.edu	**Fax:**	FY 847 491-2178
PI Organization Type:	UNIVERSITY	**Phone:**	847 491-5397
Organization Name:	Northwestern University		
PI Address 1:	Department of Engineering Sciences and Applied Mathematics		
PI Address 2:			
PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60208	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/15/2000	**End Date:**	06/16/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
Contact Email:	walter.m.duval@nasa.gov		
Flight Program:			
Flight Assignment:	note - end date changed from 11/30/2003		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2369		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Nonlinear stability theory, asymptotic and numerical methods are used to investigate the stability of coupled systems describing the directional solidification of binary systems from the melt. This work concerns the effort to understand on a quantitative level how various factors affect the morphology of solidification fronts, and hence the resulting microstructres of the crystalline solid. The project aims at the theoretical prediction of microstructure in crystalline materials which would allow the a priori design of new materials. We aim at answering the questions: How and under what conditions can crystals be grown in microgravity with different and better properties than on Earth.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	We have made enormous progress on the prediction of the structure and coarsening of quantum dots. By using long-scale appoximations the power laws of coarsening and their mechanisms have been determined for crystals possessing strong anisotropy.
Bibliography Type:	Description: (Last Updated: 10/05/2009)
Articles in Peer-reviewed Journals	Norris SA, Watson SJ. "Geometric simulation and surface statistics of coarsening faceted surfaces." Acta Materialia 2007 Nov;55(19):6444-52. http://dx.doi.org/10.1016/j.actamat.2007.08.014 , Nov-2007
Books/Book Chapters	Watson SJ. "Crystal Growth, Coarsening and the Convective Cahn-Hilliard Equation." in "Free Boundary Problems." Ed. P. Colli, C. Verdi, A. Visintin. Springer-Birkhäuser, 2004. International Series of Numerical Mathematics , Vol. 147., May-2004